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$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -x_1 + x_1 x_2 \\ x_2 - x_1 x_2 \end{bmatrix}, \quad x(0) = x_0 \in \mathbb{R}^2$$

$$\frac{dx_1}{dt} = -x_1(1-x_2)$$

$$\frac{dx_2}{dt} = x_2(1-x_1)$$

$$\|f(x,t) - f(y,t)\| = \left\| \begin{pmatrix} -x_1 + x_1 x_2 + y_1 - y_1 y_2 \\ x_2 - x_1 x_2 - y_2 + y_1 y_2 \end{pmatrix} \right\|$$

Using the 1-norm (tried first using 2-norm, but couldn't reach anywhere),

$$\|f(x,t) - f(y,t)\| = |x_1(x_2-1) - y_1(y_2-1)| + |x_2(1-x_1) - y_2(1-y_1)|$$

We need to compare it with $\|x-y\|$ to apply the Lipschitz condition,

$$\|x-y\| = \|(x_1-y_1, x_2-y_2)\| = |x_1-y_1| + |x_2-y_2|$$

$$\begin{aligned} \|f(x,t) - f(y,t)\| &= |y_1 - x_1 + x_1 x_2 - y_1 y_2| + |x_2 - y_2 + y_1 y_2 - x_1 x_2| \\ &\leq \|x-y\| + 2|x_1 x_2 - y_1 y_2| \quad \text{--- } \{ \text{Triangle Inequality} \} \\ &= \|x-y\| + 2|x_1(x_2 - y_2) + y_2(x_1 - y_1)| \\ &\leq \|x-y\| + 2\{ |x_1| |x_2 - y_2| + |y_2| |x_1 - y_1| \} \end{aligned}$$

We want local Lipschitz continuity around x_0 .

For every compact $D \subset \mathbb{R}^2$, $\exists L > 0$ such that

$$\|f(x,t) - f(y,t)\| \leq L \|x-y\| \quad \forall x,y \in D.$$

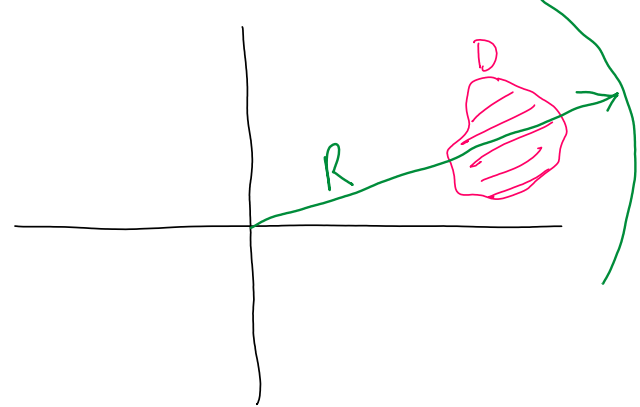
Let $D = \bar{B}_R(0)$
closed ball of radius R about $0 = (0,0)$

Thus, $|x_1| \leq R$ & $|y_2| \leq R$

$$\begin{aligned} \text{Hence, } \|f(x,t) - f(y,t)\| &\leq \|x-y\| + 2R \|x-y\| \\ &\leq (1+2R) \|x-y\| \end{aligned}$$

Take $L = 1+2R$ for this compact subset

Any other compact subset of $\mathbb{R}^2$ can be enclosed by such a ball and that L value can then be used.



Thus, f is locally Lipschitz, and this ensures

a unique solution on $[0, T)$ for some $T > 0$

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$$\ddot{y}(t) + y(t) + (2+\sin t)y(t) = 0$$

$$x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\rightarrow \dot{x} = \begin{bmatrix} x_2 \\ -x_1 - (2+\sin t)x_1 \end{bmatrix}$$

$$\text{Take } V(x,t) = \frac{x_1^2}{2} + \frac{x_2^2}{2(2+\sin t)} \gg \left\{ \frac{x_1^2}{2} + \frac{x_2^2}{2} \right\} \frac{1}{3} \gg \frac{\|x\|^2}{6} \quad \text{Thus, positive definite}$$

$$\text{Also, } V(x,t) \leq \frac{x_1^2}{2} + \frac{x_2^2}{2} \leq \frac{\|x\|^2}{2} \quad \text{Thus, decrescent}$$

$$\begin{aligned} \dot{V} &= x_1 x_2 + \frac{1}{2} \left\{ \frac{2x_2(-x_2 - (2+\sin t)x_1)}{(2+\sin t)} - \frac{x_2^2(\cos t)}{(2+\sin t)^2} \right\} \\ &= \left\{ x_1 x_2 \cdot 2(2+\sin t)^2 + \left\{ -2x_2^2(2+\sin t) - 2x_1 x_2(2+\sin t)^2 - x_2^2 \cos t \right\} \times \frac{1}{(2+\sin t)^2} \right\} \\ &= \frac{-x_2^2}{(2+\sin t)^2} (4 + 2\sin t + \cos t) \leq 0 \end{aligned}$$

Thus, negative semidefinite.

The system is uniformly stable.

9.3] $\ddot{y}(t) + p(t)\dot{y}(t) + e^{-t}y(t) = 0$

$$x = \begin{bmatrix} y \\ \dot{y} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} x_2 \\ -p(t)x_2 - e^{-t}x_1 \end{bmatrix}$$

$$V(x,t) = e^t \frac{x_1^2}{2} + \frac{x_2^2}{2}$$

$$\begin{aligned} \dot{V} &= x_1 x_2 e^{-t} - \frac{e^{-t} x_2^2}{2} - p(t) x_2^2 - e^{-t} x_1 x_2 \\ &= -\left\{ \frac{e^{-t} x_2^2}{2} + p(t) x_2^2 \right\} \end{aligned}$$

$p(t)$ being a class K function is a sufficient condition for the system to be stable.

9.4] $\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_1(1-x_1^2-x_2^2) - x_2 \\ x_2(1-x_1^2-x_2^2) + x_1 \end{bmatrix}$

The set $x_1^2 + x_2^2 = 1$ is the compact invariant set which is asymptotically stable.

Origin is unstable because trajectories spiral the unit circle no matter where the initial point is.

9.5] $\dot{x} = \begin{bmatrix} -2x_1 + x_1 u \\ -x_1 u \end{bmatrix}$

(a) Let $V(x,t) = \frac{x_1^2}{2} + \frac{x_2^2}{2}$
 $\Rightarrow \dot{V} = -2x_1^2 + x_1 x_2 u - x_1 x_2 u$
 $= -2x_1^2 \leq 0$

(b) $\dot{V}(0+) = 0 \quad \& \quad V \leq 0$
 Also, $\dot{V}$ is negative semi-definite

(c) V is non increasing and bounded below by

Thus, $\lim_{t \rightarrow \infty} V$ exists & is finite

V is bounded $\Rightarrow x_1$ & x_2 are bounded
 $\dot{V} = -2x_1^2$

$\Rightarrow \int_0^\infty \dot{V} dt = -2 \int_0^\infty x_1^2 dt$
 $\Rightarrow V(\infty) - V(0) = -2 \|x_1\|_2^2$
 $\therefore x_1 \in L_2$

$\dot{x}_1 = -2x_1 + x_1 u \in L_\infty$

Also, using Barbalat's lemma, $x_1 \rightarrow 0$

Nothing can be said about x_2 , since by

proceeding with the integral chasing, we'll

have $u x_1 \rightarrow 0$

9.6] $\begin{matrix} \dot{e}_1 = e_2 \\ \dot{e}_2 = -k_1(t)e_1 - k_2(t)e_2 \end{matrix}$ Let K & k_i be bounded

$V = \frac{k_1(t)e_1^2}{2} + \frac{e_2^2}{2} \quad \left\{ \text{Assume } k_1 > 0 \right\}$
 $\Rightarrow \dot{V} = k_1 e_1 e_2 - k_1 e_1^2 - k_2 e_2^2 + e_2^2 k_2$
 $= -k_1 e_1^2 + \frac{e_2^2 k_2}{2}$

For $\dot{V} \leq 0$, we can assume $k_2 \geq 0$ & $k_1 \leq 0$

Thus, $\lim_{t \rightarrow \infty} V(t) < \infty \Rightarrow V \in L_\infty \Rightarrow e_1, e_2 \in L_\infty$

Thus, $\dot{V} = -k_1 e_1^2 + \frac{k_2 e_2^2}{2} \leq -k_1 e_1^2$
 Let $k_1(t) \geq \alpha > 0$
 Thus, $\dot{V} \leq -\alpha e_1^2$

$\int_0^\infty \dot{V} = V(0) - V(\infty) \leq -\alpha \|e_1\|_2^2$
 Thus, $e_1 \in L_2$

$e_2 = -k_1 e_1 - k_2 e_2 \Rightarrow$ bounded

$$\lim_{t \rightarrow \infty} e_1 \in L^\infty$$

$$\lim_{t \rightarrow \infty} e_2 \rightarrow 0$$

$$\int_0^1 \dot{e}_2 = -e_2(0)$$

$$\lim_{t \rightarrow \infty} e_2 \text{ is integrable}$$

$$\begin{aligned} \dot{e}_2 &= -k_1 e_1 - k_2 e_2 - k_3 e_2 - k_4 e_2 \\ &= -k_1 e_1 - k_2 e_2 - k_3 e_2 - k_4 (-k_1 e_1 - k_2 e_2) \end{aligned}$$

$$\text{Assuming boundedness of } k_1, k_2$$

$$\text{we have } e_2 \in L^\infty \{ \text{and } e_1, e_2 \in L^\infty \}$$

$$\lim_{t \rightarrow \infty} e_2 \text{ is uniformly continuous.}$$

$$\lim_{t \rightarrow \infty} e_2 \rightarrow 0$$

$$\dot{e}_2 = -k_1 e_1 - k_2 e_2$$

$$\text{as } t \rightarrow \infty, e_1 \rightarrow 0 \text{ \& } e_2 \rightarrow 0$$

$$\text{Assuming } k_1 \geq \beta > 0, e_1 \rightarrow 0$$